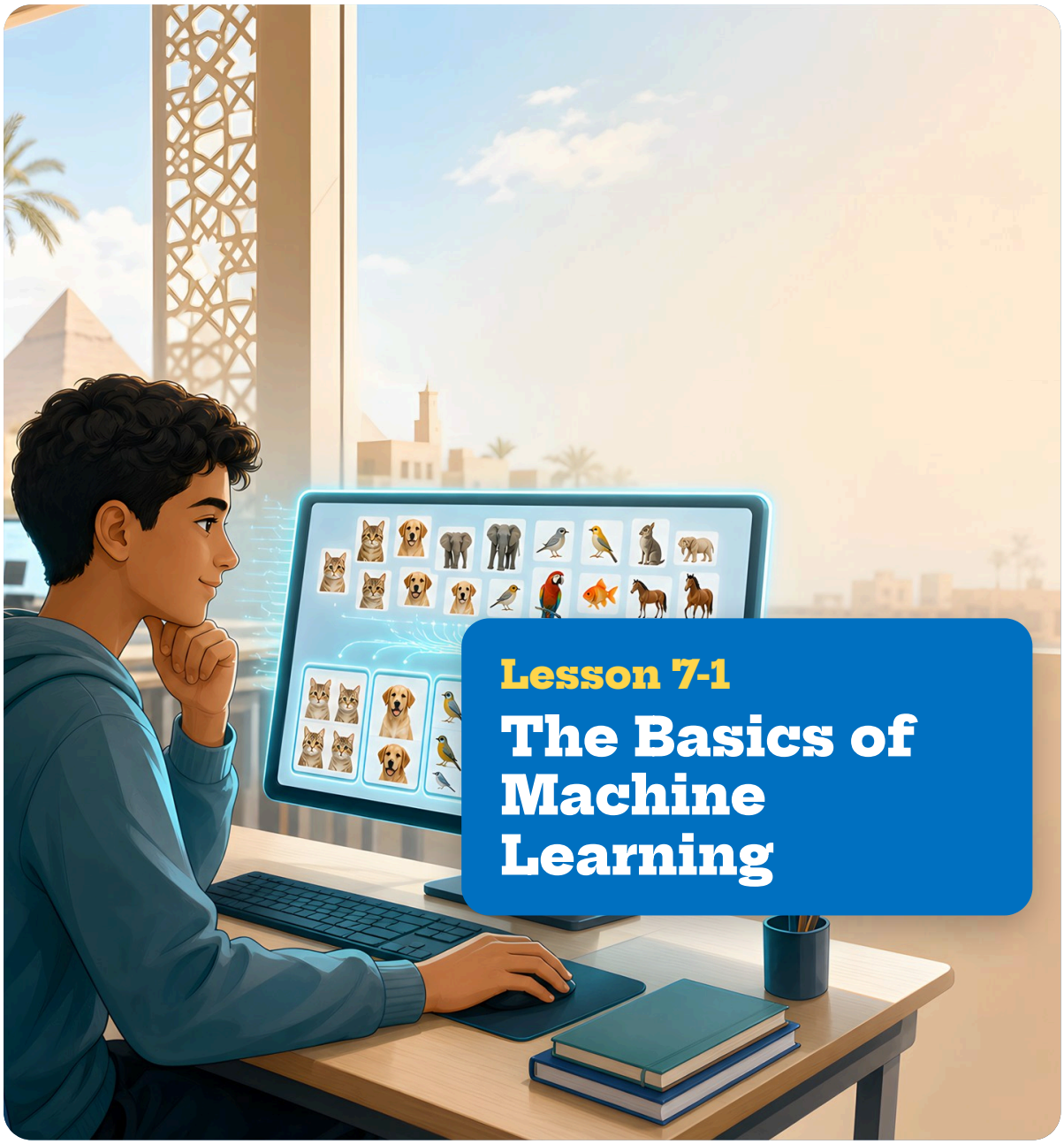


Lesson 7-1

The Basics of Machine Learning

Learning Objectives

- Explain the idea of machine learning and distinguish supervised, unsupervised, and reinforcement learning.
- Describe classification, clustering, and reinforcement learning with examples, and explain the limits caused by data bias and the black-box problem.

How do you teach a computer to tell a cat from a dog when you cannot write down a rule for it? You do not write the rule — you show it thousands of examples and let it find the rule itself. That is machine learning, the engine behind most of today's AI. It learns patterns from data to make judgments on new data. Yet it is not magic: if the examples are biased, the judgments are biased, and its reasoning can be a black box even to its makers. Machine learning is therefore best used to support human judgment, not to replace it.

? Guiding question: How does machine learning solve problems differently from ordinary programming, and what are its main types and limits?

Point!

1. The Idea of Machine Learning

(1) **(Machine learning)**…a technology that learns patterns and regularities from large amounts of data so that judgments and predictions can be made on new data. It is the most common method used to make AI work.

(2) Conventional programming and machine learning differ greatly in how they solve problems.

Approach	What humans provide	What the computer does
Conventional programming	Decision rules (procedures) and input data	Produces answers by following the rules
Machine learning	Large amounts of data (with correct answers, where available)	Finds the decision rules from the data on its own

(3) Machine learning is used for problems where it is difficult for humans to express the rules in words. *e.g.,* judging whether a photo shows a dog or a cat; distinguishing whether an email is spam; transcribing audio into text

(4) For such problems, the rules are too complex for humans to write out completely, so the approach of letting the computer learn from examples (data) becomes effective.

(5) However, machine learning is not a panacea. Because the quality and bias of the training data affect the results, biased data leads to biased judgments (bias). There is also the black-box problem, in which the basis for the model's judgments becomes hard for humans to see. Given these limitations, it is important to use machine learning not as a replacement for human judgment, but as a tool that supports human judgment.

2. Three Types of Learning Methods

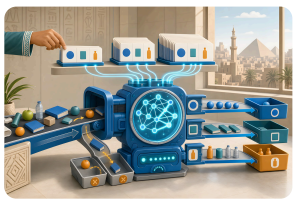
(1) Machine learning is broadly divided into the following three types, depending on the form of data given and the method of learning.

Key Fact:
Machine learning finds the rules from data instead of being given them. It comes in three forms — supervised (learn from labeled examples), unsupervised (find hidden groups), and reinforcement (learn from rewards) — but its results depend on the data, so bias and the black-box problem make it a support for human judgment, not a replacement.

Key Concepts:

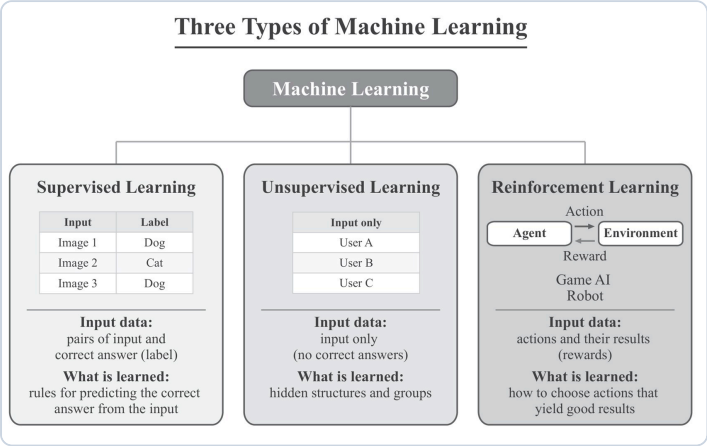
- machine learning
- supervised / unsupervised / reinforcement
- classification, feature
- clustering
- bias, black-box problem

KEY TERMS
machine learning — learning rules from data to judge new data.
bias — skewed data leading to skewed judgments.
black-box problem — the basis of a judgment is hard to see.



Machine learning improves from more examples.

Learning method	Data given	What is learned
(Supervised learning)	Pairs of input and the correct answer (label)	Rules for predicting the correct answer from the input
(Unsupervised learning)	Input only (no correct answers)	Hidden structure or groups in the data
(Reinforcement learning)	Actions and their results (rewards)	How to choose actions to obtain better results



The three types of machine learning and the data each uses

(2) Which learning method to use is determined by the type of data at hand and the nature of the problem to be solved.

Situation	Suitable learning method
Past data has correct answers, and you want to predict the correct answer for new data	Supervised learning
Data is available but without correct answers, and you want to find tendencies or groups in the data	Unsupervised learning
You want the system to learn how to choose actions for better results through trial and error	Reinforcement learning

KEY TERMS
unsupervised learning — finds groups with no labels.
reinforcement learning — learns actions from rewards.
label — the known correct answer on training data.

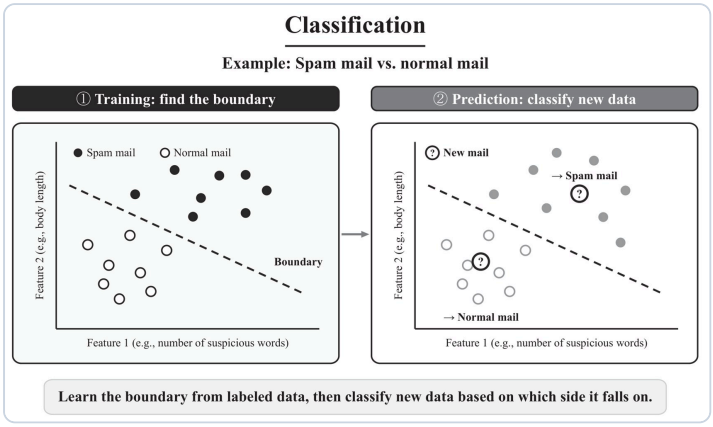
3. Classification (A Representative Example of Supervised Learning)

(1) **(Classification)**…a method of predicting which type (class) the input data belongs to. It is a representative example of supervised learning. *e.g.,* sorting emails into “spam” or “normal”; sorting medical images into “abnormal” or “normal”; classifying animals in photos as “dog,” “cat,” or “bird”

(2) In classification, large amounts of data with correct answers (labels) attached are given in advance for learning. After training is complete, the model can predict which class new data without known labels belongs to.

KEY TERMS
supervised learning — learns from labeled examples.
classification — predict which class an input belongs to.
feature — a clue in the data useful for prediction.

Stage	What is done
(Training)	Find rules for classification from pairs of input data and correct answers (labels)
(Prediction)	Use the learned rules to determine which class new data belongs to



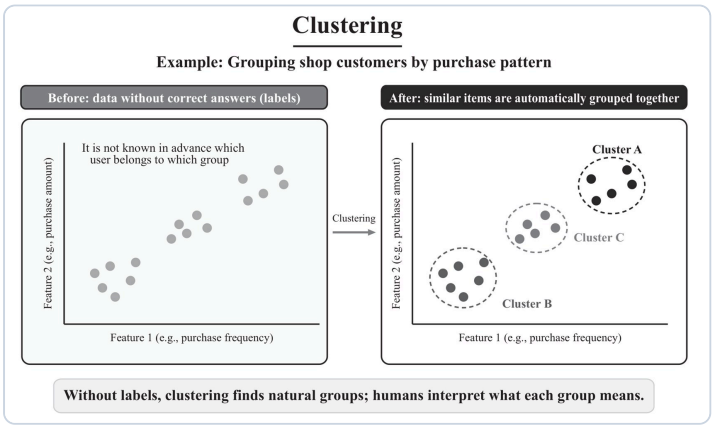
Classification: learn a boundary from labeled data, then classify new data

(3) Information in the training data that serves as a clue useful for prediction is called a **(feature)**. *e.g.,* In spam classification, the types of words contained in the email, the sender's address, and the length of the body are features.

(4) The choice of features greatly affects the accuracy of classification. Choosing features that match the purpose is important for building a good classification model.

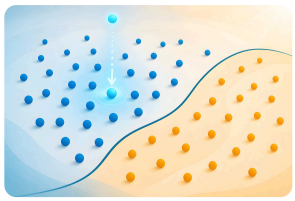
4. Clustering (A Representative Example of Unsupervised Learning)

(1) **(Clustering)**…a method that groups similar items together into groups (clusters) within data. It is a representative example of unsupervised learning. *e.g.,* grouping users of an online shop by similar purchase tendencies; grouping users with similar listening patterns on a music streaming service for recommendations; automatically grouping news articles with similar content



Clustering: without labels, similar items are automatically grouped

(2) The difference between clustering and classification lies in whether the correct answers (labels) are given.



Classification draws a boundary between labeled groups.

PAUSE & THINK

Labels given or not given — which one word separates classification from clustering?

Method	Correct answers (labels)	Purpose
Classification	Given in advance	Predict which class new data belongs to
Clustering	Not given	Find natural groups within the data

(3) In clustering, the computer itself finds "what is similar." Even when the names or number of groups are not known in advance, grouping can be done automatically based on the features of the data.

(4) Clustering is used when one wants to grasp the overall picture of the data or when one wants to look at the data from a new perspective. Humans interpret the results of grouping and use them in marketing, product development, and so on.

5. Reinforcement Learning

(1) **(Reinforcement learning)**…a method in which actions are chosen in a certain situation, and how to choose better actions is learned based on the reward obtained as a result. *e.g.,* learning how to obtain a high score in a game through trial and error; a robot learning to walk without falling by trying movements; a self-driving car learning safe driving operations

(2) In reinforcement learning, learning is accomplished by repeatedly trying combinations of actions and results.

Element	Description
(State)	The situation in which the agent (the learning side) is placed
(Action)	The choices that can be selected in that situation
(Reward)	The evaluation given as a result of the action (a numerical value indicating goodness or badness)

(3) What distinguishes reinforcement learning from the other two learning types is that it learns from the consequences of its own actions. Unlike supervised learning, where there is a correspondence of "this input has this correct answer," reinforcement learning gradually learns better choices from the rewards obtained for the actions it tries.

(4) Reinforcement learning is suited to problems where the correct answer is not known in advance, but the goodness of the result can be judged. It is often used in combination with environments where trial and error can be repeated (simulators or games).

KEY TERMS

clustering — group similar items with no labels.
reinforcement learning — learn actions from rewards.
reward — a value scoring how good an action was.



Reinforcement learning learns by reward and penalty.

PAUSE & THINK

Learning from rewards through trial and error — which of the three types is that?

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A Machine learning is a technology that learns patterns and regularities from large amounts of data so that judgments and predictions can be made on new data.

B In supervised learning, the model learns from data without correct answers (labels).

C Clustering is a method that groups data according to labels given in advance.

D Reinforcement learning is a method in which how to choose better actions is learned based on the reward obtained as a result of an action.

(2) Match each description (a–c) with the most appropriate term from the choices below (A–C).

a A learning method that learns rules for predicting correct answers for inputs from pairs of input and correct answer

b A method that finds groups of similar items from data without correct answers

c Information in the input data that serves as a clue for prediction in classification

[Choices] **A** Feature **B** Supervised learning **C** Clustering

✓ Solution

(1)

A Correct definition of machine learning. Therefore, ○.

B Data without correct answers (labels) is unsupervised learning. In supervised learning, correct answers are given. Therefore, ×.

C Clustering is a method that finds natural groups from data without given labels. Therefore, ×.

D Correct description of reinforcement learning. Therefore, ○.

(2)

a: Learning from pairs of input and correct answer is supervised learning. **B**

b: Finding groups from unlabeled data is clustering. **C**

c: Information that serves as a clue for prediction is a feature. **A**

Try

1 Answer the following questions.

1. What is the term for the technology that learns patterns and regularities from large amounts of data so that judgments and predictions can be made on new data?
2. In machine learning, what is the term for the information in the input data that serves as a clue useful for prediction?
3. What is the term for the representative method of unsupervised learning that groups similar items together?
4. In reinforcement learning, what is the term for the numerical value that represents the evaluation given as a result of an action?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that is NOT an appropriate situation in which machine learning is used.

A Distinguishing the type of animal in a photograph.

B Transcribing audio into text.

C Calculating the sum of two integers.

D Distinguishing whether an email is spam.

(2) For each of purposes (a–c), choose the most appropriate learning method from the choices below (A–C).

a Predict, from past purchase history data and whether or not a purchase was made, whether a new customer will buy the product.

b Find groups of customers with similar purchase tendencies from customer purchase data.

c Have a robot walk and learn movements that allow it to advance without falling through trial and error.

[Choices] **A** Unsupervised learning **B** Supervised learning **C** Reinforcement learning

EXAM-STYLE QUESTION

Explain why machine learning should support human judgment rather than replace it. **[6] Refer to:** bias in data and the black-box problem.

PAUSE & THINK

Which task has a simple rule a human can already write — so needs no learning?

(3) From the following options (A–D), choose the one that most appropriately describes the difference between classification and clustering.

A Classification divides data into a small number of groups, and clustering divides data into many groups.

B Classification handles only numerical data, and clustering handles only text data.

C Classification learns from data with correct answers (labels), and clustering learns from data without correct answers.

D Classification is unsupervised learning, and clustering is supervised learning.

(4) From the following options (A–D), choose the one that is NOT an appropriate point of caution when using machine learning.

A If the training data is biased, the model's judgments may also be biased.

B The basis for a machine learning model's judgments may be hard to see, so for important judgments human checking is needed.

C Machine learning is a panacea and can make more accurate judgments than humans on any problem.

D It is important to use machine learning as a tool that supports human judgment.

⚙ Think as an Engineer — Investigate & Decide

Your school has years of data and wants help with two tasks: (i) flagging students who may need extra support early, and (ii) discovering natural groups of study habits it did not know about. You advise on how to use machine learning responsibly.

☐ Let the model decide alone **☐ Choose the right method, and keep a human in charge**

1 · Identify the method. For task (i) and task (ii), name which learning type fits (supervised, unsupervised, or reinforcement). For each, name one piece of data the school would need to collect, and one kind of student that data might miss.

2 · Analyze the risks. Name one way the training data for task (i) could be biased (for example, if it only includes students who already asked for help), and say what harm the black-box problem could cause if a teacher acts on a flag without understanding it — a judgment the lesson's tables do not make for you.

3 · Decide the role of the human. State which decisions you would keep a human in charge of, and which you would let the model assist with, and why. **In pairs,** compare plans and agree on the one safeguard you would never remove.

Give evidence for your answer.

KEY TERMS

label — the known correct answer attached to data.

human-in-the-loop — a person checks or approves the model's output.

Stuck? Start from the data you have: are there correct answers to learn from (supervised) or none (unsupervised)? Then ask, for each decision, what it would cost to be wrong — the higher that cost, the more a human must stay in charge.

🌍 In a New Context

A bank wants software that reads each loan application and flags the ones that look risky for a human officer to review. Name which learning type this is, name one feature the model might use, and explain why the officer must stay in the loop rather than letting the model decide alone.

❓ Lesson Question — Answered

The lesson asked how machine learning differs from ordinary programming and what its main types and limits are. In conventional programming, humans provide the rules and the computer follows them; in machine learning, humans provide data and examples, and the computer finds the rules itself — which is why it suits problems whose rules are too complex to write, like recognizing images or speech. It comes in three types: supervised learning (labeled examples, whose representative task is classification, using features to predict a class), unsupervised learning (no labels, whose representative task is clustering, finding natural groups), and reinforcement learning (learning from rewards through trial and error, described by state, action, and reward). Its limits matter as much as its power: results depend on the quality and bias of the data, so biased data gives biased judgments, and the black-box problem hides the basis of a decision. For these reasons machine learning is best used to support human judgment, not to replace it.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

Machine learning is a technology that learns patterns and regularities from large amounts of data so that judgments and predictions can be made on new data. Machine learning is broadly divided into three types depending on the form of data given and the method of learning. There is (a) learning, which learns from pairs of input and correct answer; (b) learning, which finds structure or groups from data without correct answers; and (c) learning, which learns how to choose better actions based on the (d) obtained as a result of an action. In (e), a representative example of supervised learning, the choice of (f), which serves as a clue for prediction, greatly affects the result. In (g), a representative example of unsupervised learning, the computer itself finds "what is similar" and discovers natural groups.

1. Fill in the blanks (a)–(g).

3 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes the difference between conventional programming and machine learning.

A In conventional programming, humans provide the decision rules; in machine learning, the computer finds the decision rules from data on its own.

B Conventional programming requires large amounts of data, and machine learning requires only rules.

C Conventional programming is included in AI, but machine learning is not included in AI.

D Conventional programming can handle new problems, but machine learning can handle only fixed problems.

(2) For each of situations (a–c), choose the most appropriate term from the choices below (A–C).

a There are large amounts of news article texts and their categories (politics, economics, sports, etc.), and you want to automatically determine the category of new articles.

b You have records of products purchased by users of an online shop and want to find groups of users with similar purchasing tendencies.

c You want a computer to play a game and learn the operations that lead to high scores through trial and error.

[Choices] **A** Reinforcement learning **B** Clustering **C** Classification

REFLECT & CHALLENGE

Reflect: Where have you likely met machine learning today (recommendations, photo tagging, voice input)? Which type do you think it was?
Challenge: A face-recognition model was trained mostly on one group of people and works poorly on others. Explain how biased training data caused this, and state one change to the data that would reduce the bias.

Key takeaway

Remember:
Machine learning finds rules from data: supervised (labeled → classification with features), unsupervised (unlabeled → clustering), reinforcement (rewards). Mind its limits — bias and the black-box problem — and keep it supporting human judgment.

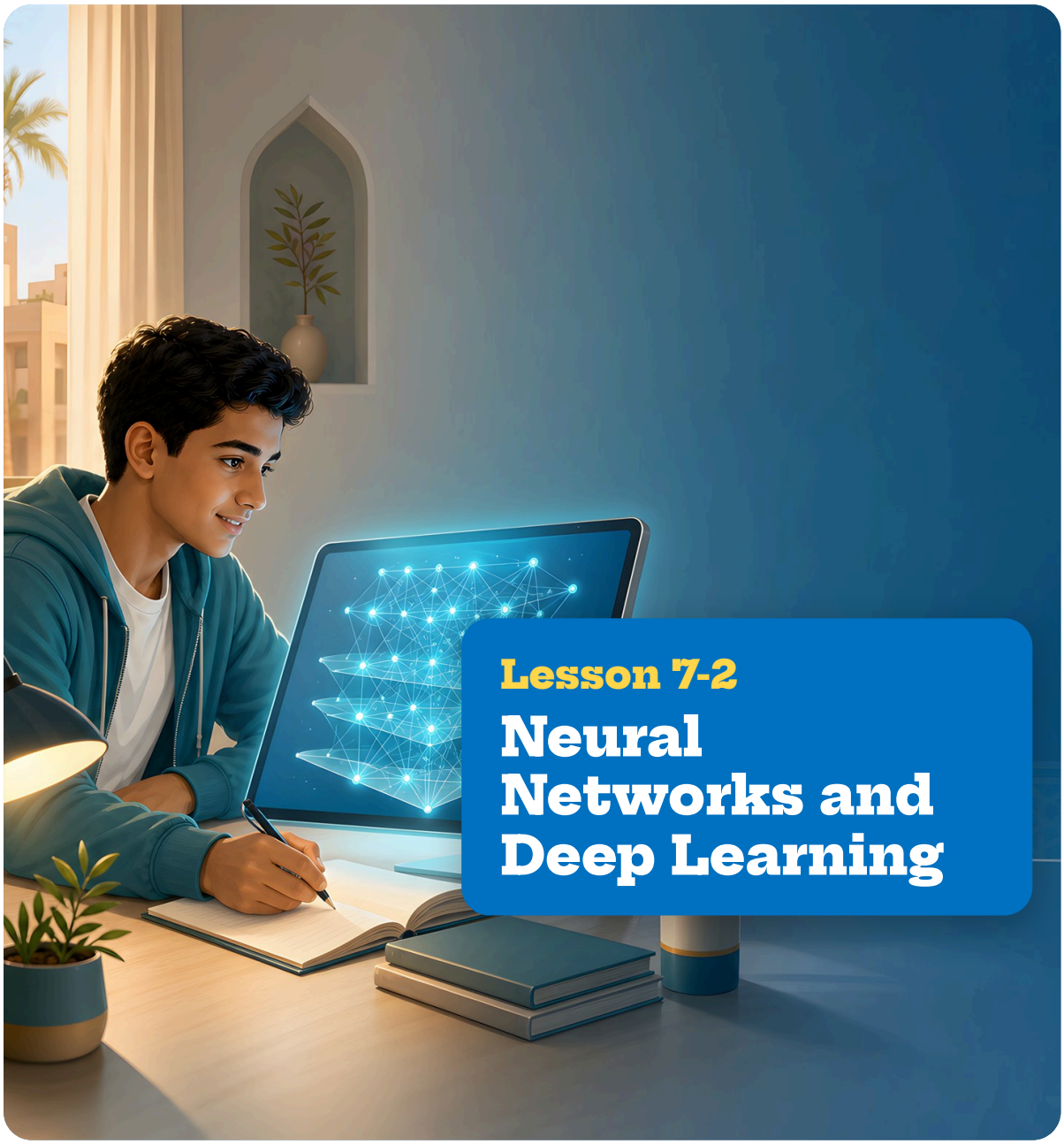
(3) From the following options (A–D), choose the one that is NOT an appropriate description of the characteristics and limitations of machine learning.

A Machine learning is used for problems where it is difficult for humans to express the rules in words.

B Machine learning models are affected in their results by the quality and bias of the training data.

C A machine learning model's basis for judgment is always shown to humans in an easy-to-understand way.

D It is important to use machine learning not as a replacement for human judgment, but as a tool that supports human judgment.



Lesson 7-2

Neural Networks and Deep Learning

Learning Objectives

- Describe the structure of a neural network and explain how a single neuron processes its inputs.
- Explain what deep learning is, and why stacking many hidden layers brought a leap in AI performance.

How can a machine tell a dog from a cat, or turn speech into text? The answer, loosely inspired by the brain, is a neural network: layers of simple units called neurons, each passing signals to the next. A neuron just multiplies its inputs by weights and adds them up — nothing complex on its own — but stack enough of them across enough layers and the network learns, by adjusting those weights on data, to recognize patterns no human could write rules for. Making the stack deep — deep learning — lets the network build up features from simple edges up to whole objects, and it is what pushed AI past humans on many tasks.

? Guiding question: How is a neural network built and trained, and what does making it "deep" add?

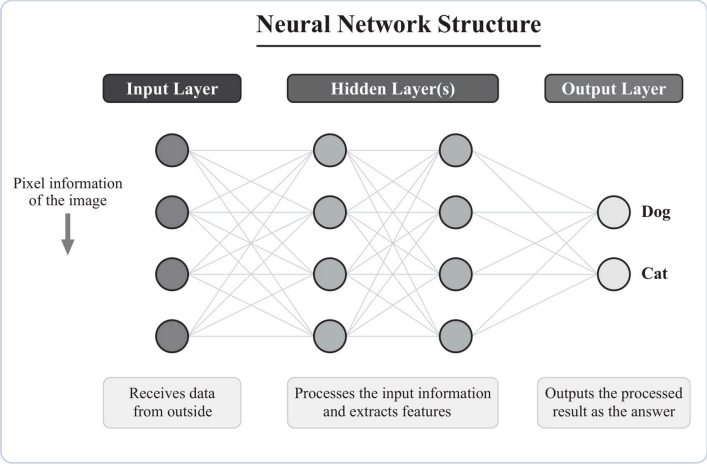
Point!

1. The Structure of Neural Networks

(1) A neural network has a structure in which many **(nodes)**, also called neurons, are connected. Nodes are arranged together in **(layers)**, and the nodes of adjacent layers are connected to each other.

(2) Layers are divided by role into the following three types.

Type of layer	Role
Input layer	Receives data from outside
Hidden layer (middle layer)	Processes the input information and extracts features
Output layer	Outputs the result of processing as an answer



A neural network: input layer, hidden layers, and output layer

(3) Data enters at the input layer, is processed as it passes through the hidden layers, and reaches the output layer in this flow. *e.g.,* In a network that judges whether the animal in a photograph is a dog or a cat, the input layer receives the

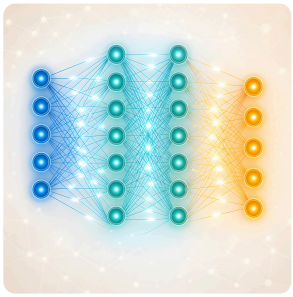
Key Fact:
A neural network is layers of connected neurons: an input layer, hidden layers that extract features, and an output layer. Each neuron multiplies its inputs by weights and sums them; training adjusts those weights from data. Deep learning stacks many hidden layers to build features step by step — the leap behind modern AI.

Key Concepts:

- node (neuron), layer
- input / hidden / output layer
- weight, training
- deep learning
- hierarchical features

KEY TERMS

node (neuron) — a single processing unit.
layer — a group of neurons at one stage.
input / hidden / output layer — receive, process, answer.



A network stacks neurons into connected layers.

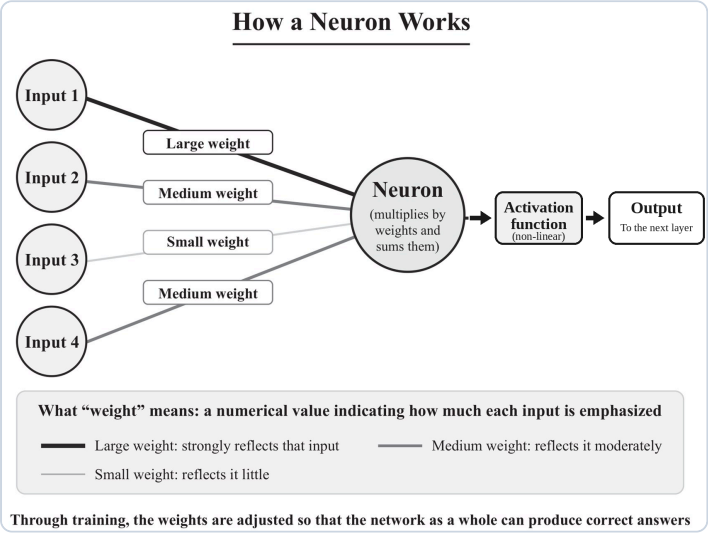
pixel information of the photograph, the hidden layers extract features such as shape and pattern, and the output layer outputs the judgment of “dog” or “cat.”

(4) The number of hidden layers and the number of nodes in each layer are designed according to the difficulty of the problem to be handled. The more complex the problem, the more layers and nodes are needed.

2. How a Neuron Works

(1) Each neuron receives multiple inputs, multiplies them by weights and sums them, then passes the sum through an **(activation function)**, which lets the network learn relationships that are not simply proportional, before passing the result to the next layer.

(2) **(Weight)**…a numerical value that represents how much each input is emphasized. Weights can be negative as well as positive, and the larger a weight is in size, the more strongly that input influences the neuron's result.



A neuron weights its inputs, sums them, applies an activation function, and passes the result on

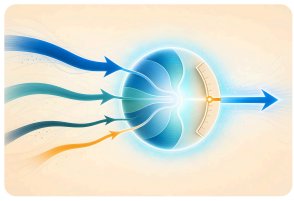
(3) **(Training)** is the process of gradually adjusting the weights between neurons using a large amount of data.

Stage	What happens
Before training	The weights are set to arbitrary values, and correct predictions cannot be made
During training	Data is given, and the weights are gradually adjusted based on the difference between the prediction and the correct answer
After training	With the adjusted weights, correct predictions can be made even on new data

(4) As training progresses, a combination of weights that allows the network as a whole to produce correct answers is found. Even though the function of each individual neuron is simple, complex judgments become possible by combining many neurons.

KEY TERMS

weight — how much an input is emphasized.
training — adjusting weights from data.



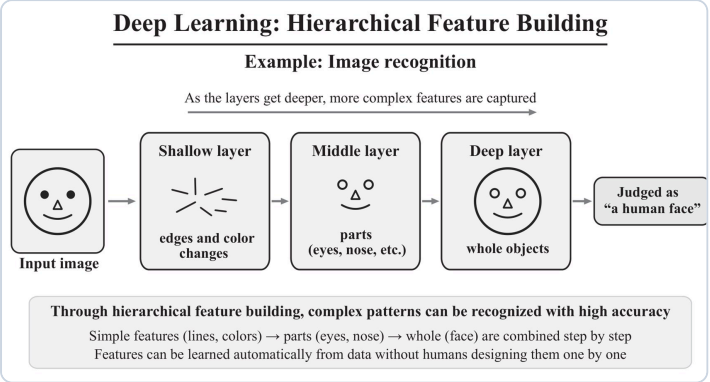
A neuron weights its inputs, sums them, applies an activation function, and passes the result on.

3. Deep Learning

(1) **(Deep learning)** is a technology that uses neural networks with many hidden layers stacked deeply. “Deep” refers to having many hidden layers.

(2) When the hidden layers are made deeper, features of data can be built up step by step and captured. *e.g.,* In an image-recognition network, features are built up from shallow to deep layers as follows.

Layer position	Examples of features extracted
Shallow layers	Simple features such as lines and changes in color
Middle layers	Shapes of parts such as eyes, nose, and ears
Deep layers	Structure of the whole face or object



Deep learning builds features step by step, from edges to whole objects

(3) Through this hierarchical building of features, deep learning can recognize complex patterns with high accuracy.

(4) Deep learning became practical as a result of large amounts of data and high computational power (in particular, specialized processors well suited to AI computation, such as GPUs and TPUs) becoming available. Before that, it was technically difficult to advance learning sufficiently even when layers were made deeper.

4. Why Deep Learning Brought a Leap in AI

(1) With the availability of deep learning, accuracy close to or surpassing humans has become achievable in many tasks that were previously considered difficult for computers.

(2) The fields in which there has been particularly great progress include the following.

PAUSE & THINK

Why does stacking more hidden layers let a network capture more complex features?



Deep layers build from edges to whole objects.

PAUSE & THINK

What two things had to become available before deep learning could work in practice?

Field	Examples
(Image recognition)	Face recognition systems, diagnostic support for medical images, object detection for self-driving cars
(Speech recognition)	Voice operation of smart speakers, transcription from speech, speech translation
(Natural language processing)	Machine translation, summarization of text, answering questions

(3) Before deep learning, features had to be designed by humans one by one and given to the computer. With deep learning, features can be learned automatically from data, allowing the limitations of human design to be exceeded.

(4) As a result of accumulating these capabilities, the foundation has been laid for AI that newly creates text and images.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A A neural network has three types of layers: input layer, hidden layer, and output layer.

B The input layer is the layer that outputs the results of processing as an answer to the outside.

C Deep learning is a technology that uses a neural network with only one hidden layer.

D Training a neural network is the process of adjusting the weights between neurons using a large amount of data.

(2) Match each description (a–c) with the most appropriate term from the choices below (A–C).

a In a neural network, the layer that receives data from outside

b In a neural network, a numerical value that represents how much each input is emphasized

c In a neural network, the layer that processes input information and extracts features

[Choices] **A** Hidden layer **B** Weight **C** Input layer

✓ Solution

(1)

A There are three types of layers: input, hidden, and output. Therefore, ○.

B The output layer is what outputs results to the outside. The input layer is the layer that receives data from outside. Therefore, ×.

C Deep learning is a technology that uses neural networks with many hidden layers stacked deeply. Therefore, ×.

PAUSE & THINK

Which layer receives the data, and which one gives the final answer?

D Correct description of training. Therefore, ○.

(2)

a: The one that receives data is the input layer. **C**

b: The value representing how much an input is emphasized is the weight. **B**

c: The one that processes information and extracts features is the hidden layer. **A**

Try

1 Answer the following questions.

1. What is the term for the individual processing unit that makes up a neural network?
2. In a neural network, what is the term for the layer that processes input information and extracts features?
3. In a neural network, what is the term for the numerical value that represents how much each input is emphasized?
4. What is the term for the technology that uses neural networks with many hidden layers stacked deeply?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes the layers of a neural network.

A The input layer has the role of outputting processing results to the outside.

B The hidden layer has the role of receiving data from outside.

C The output layer has the role of processing input information and extracting features.

D Data enters at the input layer, is processed as it passes through the hidden layers, and reaches the output layer in this flow.

(2) From the following options (A–D), choose the one that most appropriately describes the training of a neural network.

A From before training begins, the weights between neurons are set to correct values.

B As training progresses, a combination of weights that allows the network as a whole to produce correct answers is found.

C Training is performed by increasing the number of neurons.

D Once training is complete, the network becomes unable to handle new data.

EXAM-STYLE QUESTION

Explain how a neural network with simple neurons can make complex judgments, referring to weights and training. **[6]**
Refer to: weighting inputs and adjusting weights on data.

PAUSE & THINK

Before training, are the weights already correct? What changes them?

(3) From the following options (A–D), choose the one that is NOT an appropriate characteristic of deep learning.

A It has a structure with many hidden layers stacked deeply.

B It can build up features of data step by step.

C The deeper the hidden layers, the more complex features can be captured.

D Even if the number of hidden layers is increased, the difficulty of problems that can be handled does not change.

(4) For a deep learning network that performs image recognition, reorder the following (A–C) in the order in which features are extracted from shallow layers to deep layers.

A Structure of the whole face or object

B Simple features such as lines and changes in color

C Shapes of parts such as eyes, nose, and ears

PAUSE & THINK

More hidden layers let a network capture more complex features — but what does that extra depth demand in data and computing power?

Think as an Engineer — Investigate & Decide

A team wants an app that reads handwritten digits 0–9 from scanned school registers and turns them into typed numbers. You advise on building it with a deep learning network.

☐ **Write the rules for each digit by hand**

☐ **Train a deep network on many examples**

1 · Design the layers. Say what the input layer would receive, what the output layer would produce, and in one line why hidden layers (rather than a single step) are needed here.

2 · Explain the depth. Using the idea of hierarchical features, describe what shallow, middle, and deep layers might each capture for a handwritten digit — a chain the lesson shows for faces but does not spell out for digits, so you must map it yourself.

3 · Judge the cost and risk. Name the two things this network needs to become practical, and name one way the training data could be too narrow, making the app fail on some students' handwriting. **In pairs,** compare designs and agree on the one dataset change that would most improve fairness.

Give evidence for your answer.

Stuck? Use the lesson's face example as a model, then translate it to a digit: ask what would count as an “edge,” a “part,” and a “whole” for a written number. For the risk, list whose handwriting the network saw during training — then ask whose it never saw.

KEY TERMS

training data — the labeled examples a network learns from.

generalize — work correctly on new, unseen data.

🌍 In a New Context

A voice assistant turns spoken commands into text and often works well even for words it was not explicitly programmed with. Name which field of AI progress this belongs to, and explain, using layers and training, how it can handle speech no one wrote rules for.

❓ Lesson Question — Answered

The lesson asked how a neural network is built and trained and what making it deep adds. A neural network is built from layers of connected neurons: an input layer that receives data, hidden layers that process it and extract features, and an output layer that gives the answer, with data flowing input → hidden → output. Each neuron is simple — it multiplies its inputs by weights, sums them, and passes the sum through an activation function — and the network is trained by gradually adjusting those weights on large amounts of data. Before training, arbitrary weights give wrong answers; after training, tuned weights give correct ones even on new data. Making the network deep — deep learning, many hidden layers — lets features be built hierarchically, from simple edges through parts to whole objects, so complex patterns are recognized with high accuracy. Once large data and powerful processors like GPUs made it practical, deep learning let features be learned automatically instead of designed by hand, driving leaps in image recognition, speech recognition, and natural language processing.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

A neural network has a structure in which many processing units called (a) are connected. These are arranged together in (b), and are divided into three types: (c), which receives data from outside; (d), which processes information and extracts features; and (e), which outputs the result as an answer. Each (a) multiplies its inputs by (f), sums them, passes the sum through an activation function, and passes the result to the next layer. Training gradually adjusts (f) using large amounts of data. The technology that uses neural networks with many hidden layers stacked deeply is called (g), and its characteristic is that it can build up features of data (h) and capture them.

⭐ REFLECT & CHALLENGE

Reflect: Where have you used a tool built on deep learning (voice typing, photo search, translation)? Which field was it?
Challenge: A team makes a network deeper and deeper but accuracy stops improving and training becomes very slow. Using what made deep learning practical, explain what their deeper network now demands more of, and name the one thing — besides adding layers — they should check first.

1. Fill in the blanks (a)–(h).

3 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes how a neuron in a neural network works.

A Each neuron passes one input as is to the next layer.

B Each neuron multiplies its inputs by weights, sums them, passes the sum through an activation function, and passes the result on.

C Each neuron averages the input values and passes the result to the next layer.

D Each neuron passes only the largest input to the next layer.

(2) From the following options (A–D), choose the one that most appropriately describes the background to deep learning becoming practical.

A Computation became simpler by reducing the number of hidden layers to one.

B Large amounts of data and high computational power became available.

C Connections between neurons were eliminated, simplifying processing.

D A method in which humans set weights one by one by hand was established.

(3) From the following options (A–D), choose the one that most appropriately describes why deep learning brought a leap in AI.

A Features could be learned automatically from data without humans designing them one by one.

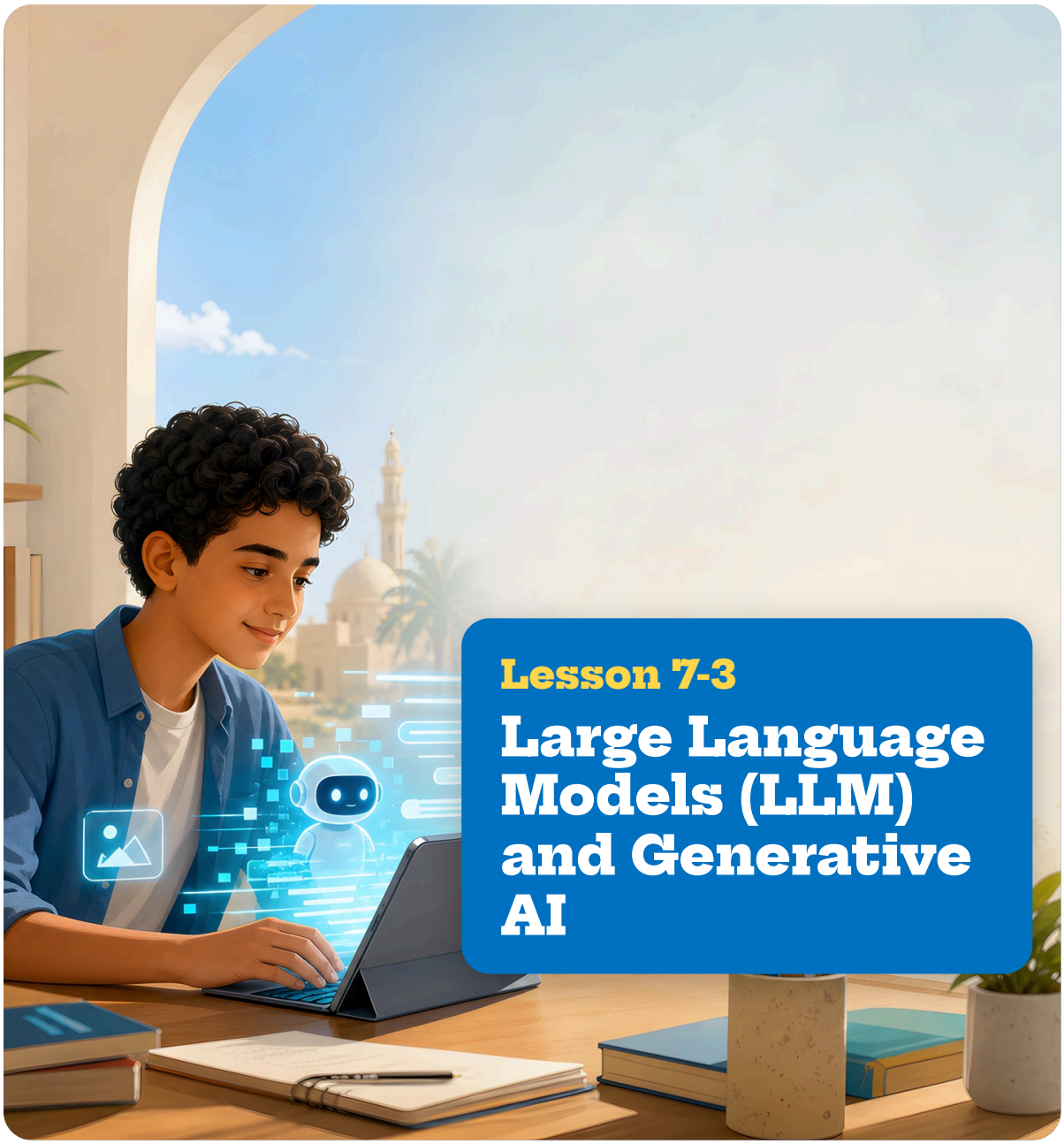
B Computers became able to think with the same mechanism as the human brain.

C Computers became able to fully understand the meaning of words.

D Computers became able to make completely accurate judgments on any problem.

Key takeaway

Remember:
A neural network = layers of neurons (input → hidden → output); each neuron weights and sums its inputs, and training tunes the weights on data. Deep learning stacks many hidden layers to build features (edges → parts → wholes), enabled by big data and GPUs.



Learning Objectives

- Explain how a language model generates text by predicting the next word, and why this causes hallucination.
- Describe what makes a language model large, and explain the roles of RLHF tuning and multimodality.

When a chatbot writes a fluent paragraph, it is not looking up an answer — it is guessing the next word, then the next, using patterns learned from enormous amounts of text. That simple trick, scaled up to billions of parameters, gives large language models their power to translate, summarize, and converse. But because each word is chosen for being statistically plausible rather than certainly true, these models can state falsehoods convincingly — a failure called hallucination. This lesson shows how such models work, how human feedback (RLHF) makes them helpful, and how multimodality lets them handle images and audio too.

? Guiding question: How does a language model generate text, what makes a model "large," and why can it still get facts wrong?

Point!

1. Language Models and "Predicting the Next Word"

(1) **(Language model)**…a model created to learn from large amounts of text and predict which word is likely to come next in a sentence. Through training, it acquires patterns of word usage and phrasing according to context.

(2) Text generation is achieved by repeating the prediction of the next word.

Step	Input sentence	Word predicted next
1st	"Today's weather is"	"sunny"
2nd	"Today's weather is sunny"	"and"
3rd	"Today's weather is sunny and"	"warm"

In this way, a complete sentence is built up by connecting the results of predicting one word at a time.

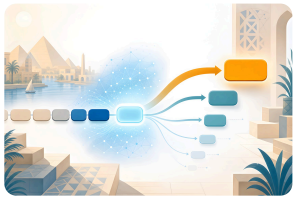
Key Fact:
A language model generates text by predicting the next word again and again — choosing what is statistically plausible, not guaranteed true (the cause of hallucination). Large language models scale this up with vast data and billions to trillions of parameters, are tuned by RLHF to be helpful, and, through multimodality, handle text, images, and audio together.

Key Concepts:

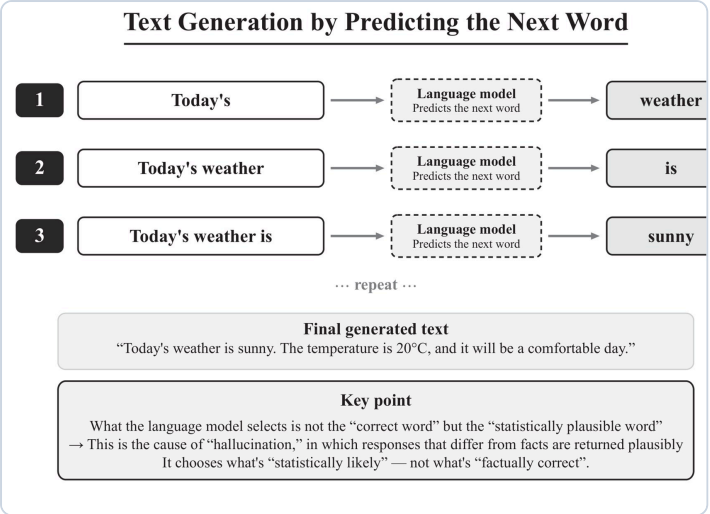
- language model, next-word prediction
- hallucination
- large language model (LLM), parameters
- RLHF
- multimodality

KEY TERMS

language model — predicts the likely next word.
hallucination — a plausible answer that is factually wrong.



A language model predicts the next word.



Text is generated by predicting the next word, one at a time

(3) For each candidate word, the language model calculates a "probability of coming next" and selects words based on that probability. These probabilities come from patterns the model learned during training, so it can also handle wordings that never appeared in the training data.

(4) What a language model selects is not the correct word, but a statistically plausible word. Because of this property, **(hallucination)**—the phenomenon of returning responses that differ from the facts but are stated in a plausible way—occurs.

2. Large Language Models (LLM)

(1) **(Large language model (LLM))**…a language model based on a very large neural network trained using vast amounts of text from the Internet.

(2) "Large" mainly refers to the following two kinds of size.

Aspect	Description
Amount of training data	The model is trained using vast amounts of text containing billions to trillions of words
Number of parameters	The numerical values corresponding to the weights of the neural network number in the billions to trillions

(3) Researchers have observed that when the number of parameters or the amount of training data exceeds a certain scale, qualitatively new abilities begin to appear. *e.g.,* handling complex reasoning; summarizing text; translating into foreign languages; generating program code

(4) Major services built on large language models include ChatGPT, Claude, and Gemini, which provide natural conversation and a wide range of text generation.

3. Tuning by RLHF

(1) Just by learning from vast amounts of text, large language models do not necessarily return responses that are desirable for humans. They may answer

KEY TERMS

large language model (LLM) — a huge network trained on vast text.

parameter — a weight; LLMs have billions to trillions.

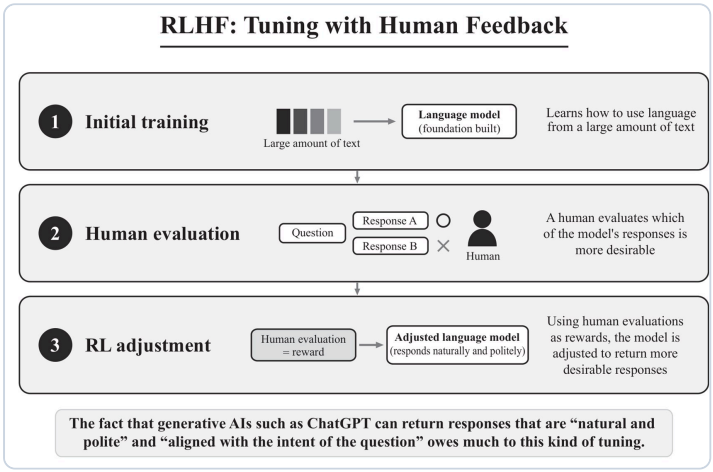
RLHF — tuning by human feedback as reward.

without grasping the intent of the question or return responses containing inappropriate content.

(2) A representative method for solving such problems is **(RLHF (Reinforcement Learning from Human Feedback))**. RLHF is a method that applies reinforcement learning to the tuning of language models.

(3) RLHF proceeds in the following flow.

Stage	Description
① **Initial training**	Build a language model that has learned word usage from large amounts of text
② **Human evaluation**	Humans give evaluations on the model's responses, such as "which is more desirable"
③ **Tuning by reinforcement learning**	Using human evaluations as rewards, tune the model to return more desirable responses



RLHF: human evaluations become rewards that tune the model

(4) Through RLHF, language models begin to return natural and polite responses, responses aligned with the intent of the question, and responses that avoid inappropriate content. The fact that generative AIs such as ChatGPT feel "easy to use" to users owes much to such tuning.

4. Multimodality

(1) **(Multimodality)**…the property of being able to handle multiple kinds of data with a single model, such as text, images, audio, and video. Each kind of data is called a **(modality)**.

(2) Early language models handled only text, but recent generative AIs can handle different kinds of data in combination. *e.g.,* looking at an image and describing its content in text; listening to audio and transcribing it into text; giving instructions in text and generating an image; summarizing the content of a video

(3) The background to multimodality being achieved is that neural networks have become able to handle different kinds of data by converting them into a

PAUSE & THINK

Left alone, a huge model may answer rudely or off-point. What step fixes that?



Human feedback ranks outputs to tune the model.

KEY TERMS

multimodality — one model handling many data kinds.
modality — one kind of data (text, image, audio…).

common numerical representation. Inside the network, text, images, and audio are all processed in the same way as sets of numerical values.

(4) Through multimodality, generative AI is evolving from a technology that handles only language into a technology that integrally handles the various kinds of information that humans deal with daily.



One model can output text, image, and sound.

PAUSE & THINK

If a model sounds confident and fluent, does that guarantee it is correct?

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A A language model is a model that predicts which word is likely to come next in a sentence.

B A language model finishes processing after predicting just one word and does not generate text.

C In a large language model, when the number of parameters or the amount of training data exceeds a certain scale, qualitatively new abilities sometimes appear.

D Multimodality is the property of handling only text.

(2) Match each description (a–c) with the most appropriate term from the choices below (A–C).

a A language model based on a very large neural network trained using vast amounts of text from the Internet

b A method that uses human feedback as a reward to tune a language model to return more desirable responses

c The phenomenon in which a language model returns responses that differ from the facts in a plausible way

[Choices] **A** Hallucination **B** Large language model **C** RLHF

✓ Solution

(1)

A Correct definition of a language model. Therefore, ○.

B Text generation is achieved by repeating the prediction of the next word. Therefore, ×.

C Correct description of large language models. Therefore, ○.

D Multimodality is the property of being able to handle multiple kinds of data—text, images, audio, video, etc.—with a single model. Therefore, ×.

(2)

a: A huge language model trained on vast text is a large language model. **B**

b: Tuning by reinforcement learning using human feedback as a reward is RLHF. **C**

c: The phenomenon of returning fact-discrepant responses in a plausible way is hallucination. **A**

Try

1 Answer the following questions.

1. What is the term for the model that learns from large amounts of text and predicts which word is likely to come next in a sentence?
2. What is the term for a language model based on a very large neural network trained using vast amounts of text from the Internet?
3. What is the term for the method that uses human feedback as a reward to tune a language model to return more desirable responses?
4. What is the term for the property of being able to handle multiple kinds of data—text, images, audio, video, and so on—with a single model?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes how a language model works.

A It memorizes the input sentence as is and returns the same sentence.

B It generates text by repeatedly predicting the next word in the sentence.

C It selects and returns the closest match from prepared template sentences.

D It fully understands the meaning of the sentence and returns the single correct response.

(2) From the following options (A–D), choose the one that most appropriately describes the cause of hallucination.

A Because the language model always retrieves and returns sentences from the training data as is.

B Because the language model selects and returns words that are statistically plausible rather than correct.

C Because the language model avoids answering questions and returns unrelated text.

D Because the language model is built to return only fact-based responses.

EXAM-STYLE QUESTION

Explain why a language model can produce a fluent answer that is factually wrong (hallucination). **[6] Refer to:** choosing statistically plausible words.

PAUSE & THINK

"Plausible" or "correct" — which one does the model actually optimize for?

(3) From the following options (A–D), choose the one that most appropriately describes RLHF.

A A method of building a language model from scratch using a large amount of text.

B A method of tuning a language model to return more desirable responses, using human evaluations as rewards.

C A method that removes the neural network structure from a language model and simplifies the processing.

D A method that has a language model solve math problems to improve its computational accuracy.

(4) From the following options (A–D), choose the one that is NOT an appropriate application of multimodal AI.

A Showing an image and having it described in text.

B Listening to audio and transcribing it into text.

C Giving instructions in text and generating an image.

D Computing the result of input numerical values using only the rules of the four arithmetic operations and returning it.

⚙ Think as an Engineer — Investigate & Decide

A school club wants to use a generative-AI chatbot to help write short articles about local history for the school website. You advise on using it responsibly.

☐ **Publish whatever the AI writes** ☐ **Use the AI, but verify before publishing**

1 · Decide where it helps. Name one part of the task where the chatbot genuinely saves effort, and one part where its output must not be trusted as-is. Explain each in one line.

2 · Analyze the risk. Using how a language model chooses words, explain why the chatbot might state a wrong date or invented name for a local landmark very confidently, and describe exactly how the club would catch such a hallucination before publishing — a check the lesson does not spell out for you.

3 · Decide the safeguards. State the rule the club should follow for every AI-written article before it goes live, and say what role RLHF-style tuning does and does not solve here. **In pairs,** compare rules and agree on the one check you would never skip.

Give evidence for your answer.

KEY TERMS

generative AI — AI that creates new text, images, or audio.

fact-check — verifying a claim against a reliable source.

Stuck? Remember the model is rewarded for *sounding* right, not for *being* right. Which items in a history article — dates, names, places — are exactly the kind of thing it could get confidently wrong? Think about who or what could confirm each one, and at what point a person should step in.

🌎 In a New Context

A student uploads a photo of a handwritten note and asks an AI assistant to read it aloud and then summarize it in text. Name the property that lets one model move between image, text, and audio, and explain in one line how the network can handle all three.

❓ Lesson Question — Answered

The lesson asked how a language model generates text, what makes a model "large," and why it can still get facts wrong. A language model learns from large amounts of text to predict the likely next word, and it generates text by repeating that prediction, connecting one word at a time into a whole sentence. It chooses each word by a probability of coming next — the statistically plausible word, not the guaranteed-correct one — which is exactly why hallucination (fluent but false responses) can occur. A large language model scales this up: trained on vast Internet text with billions to trillions of parameters, and beyond a certain scale it gains qualitatively new abilities such as reasoning, summarizing, translating, and generating code (ChatGPT, Claude, and Gemini are examples). Because raw training does not guarantee helpful answers, RLHF tunes the model with human evaluations as rewards, making responses natural, on-intent, and safer. And multimodality lets a single model handle text, images, and audio together by converting them all into a common numerical representation. The lasting caution: fluency is not truth, so outputs must still be checked.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

A model that predicts which word is likely to come next in a sentence is called a (a). Text generation is achieved by repeating the prediction of the (b). A language model based on a very large neural network trained using vast

⭐ REFLECT & CHALLENGE

Reflect: When you last used a generative-AI tool, did you check its facts? What could a confident wrong answer have cost? **Challenge:** A classmate says "RLHF makes the AI always tell the truth." Explain why that is not correct, and state what RLHF actually improves.

amounts of text from the Internet is called a (c); when the number of (d) or the amount of training data exceeds a certain scale, qualitatively new abilities sometimes appear. Since just learning from vast text does not necessarily lead to responses desirable for humans, (e), a method of tuning by reinforcement learning using human evaluations as rewards, is used. Also, the property of being able to handle multiple kinds of data—text, images, audio, video, and so on—with a single model is called (f).

1. Fill in the blanks (a)–(f).

3 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes the mechanism of text generation.

A Text is generated by connecting the results of repeatedly predicting the next word.

B Text is generated by selecting from thousands of prepared templates.

C Text is generated by first deciding the length of the sentence and then deciding all words simultaneously.

D Text is generated by machine-translating the input sentence.

(2) From the following options (A–D), choose the one that most appropriately describes the background to large language models acquiring new abilities.

A Because reducing the number of parameters below a certain level simplified the processing.

B Because limiting training data to a small amount of high-quality text raised quality.

C Because qualitatively new abilities appear when the number of parameters or the amount of training data exceeds a certain scale.

D Because humans wrote down the rules of language one by one and taught them to the model.

(3) From the following options (A–D), choose the one that most appropriately describes the purpose for which RLHF is performed.

A To increase the number of words a language model can handle.

B To make a language model's responses natural and aligned with the intent of the question.

C To enable a language model to handle data other than text.

D To reduce the number of parameters used in a language model's calculations.

⭐ Key takeaway

Remember:
A language model generates text by predicting the next word (plausible, not always true → hallucination). LLMs use vast data and billions to trillions of parameters; RLHF tunes them by human feedback; multimodality unifies text, image, and audio. Fluency ≠ truth — always verify.